

Operations Research for Health Care

Lesson 10/15

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Outline of this lecture

Modelling exercise: Data

Modelling exercise: Report

Modelling exercise: Model

Modelling exercise: Code

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Modelling exercise: Data

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Modelling exercise: Code

An hierarchical location-allocation model: Sets

Consider a hierarchical health care network with the following data. The objective is to minimise overall costs while providing universal health coverage.

- ▶ 15 demand points;
- ▶ 3 Health care levels, primary, secondary, and tertiary (PHC, SHC, THC)
- ▶ 2 Types of Patients (chronic and acute)
- ▶ 6 Existing health care units on PHC
- ▶ 4 Candidate health care location on PHC
- ▶ 4 Existing health care units on SHC
- ▶ 4 Candidate health care location on SHC
- ▶ 3 Existing health care units on THC
- ▶ 3 Candidate health care location on THC

An hierarchical location-allocation model: Sets

```
set I:= # The set of demand points.  
1      2      3      4      5      6  
7      8      9      10     11     12  
13     14     15  
;
```

```
# Health care levels (PHC, IHC, Hospital)  
set K:=  
1 # Primary care (basic care)  
2 # Secondary care (intermediate care)  
3 # Tertiary care (hospital care)  
;
```

```
# Types of Patients (chronic or acute) (Good)  
set P:=  
1 # Chronic conditions  
2 # Acute conditions  
;
```

```
# Basic health care units teams (PHC)  
set E[1] :=  
ME1 # Family physician  
EF1 # Family nurse  
ACS # Community health agent  
DE1 # Dentist  
;
```

```
# Intermediate health care units teams (SHC)  
set E[2] :=  
ME2 # Clinical physician  
EF2 # Nurse  
FON # Speech therapist  
OCD # Occupational therapist  
DE2 # Dentist  
;
```

```
# Tertiary health care units teams (Hospital)  
set E[3] :=  
ME3 # Clinical physician  
EF3 # Nurse  
TE3 # Nursing technicians  
FIS # Physiotherapist  
DE3 # Dentist  
;
```

```
# EXISTING health care units on three levels  
set EL[1] := 1 2 3 4 5 6;  
set EL[2] := 1 2 3 4;  
set EL[3] := 1 2 3;
```

```
# CANDIDADE health care LOCATIONS on three levels  
set CL[1] := 7 8 9 10;  
set CL[2] := 5 6 7 8;  
set CL[3] := 4 5 6;
```

```
# Team cost K1 ($/month)  
param CE1:=  
ME1 10  
EF1 10  
ACS 10  
DE1 10  
;
```

```
# Team cost K2 ($/month)  
param CE2:=  
ME2 20  
EF2 20  
FON 20  
OCD 20  
DE2 20  
;
```

An hierarchical location-allocation model: Data

```
# Team cost K3 ($/month)
param CE3:=
ME3 30
EF3 30
TE3 30
FIS 30
DE3 30
;

param D1
:      1      2      3      4      5      6      7      8      9      10      :=
1      1      1      1      1      1      1      1      1      1      1
2      1      1      1      1      1      1      1      1      1      1
3      1      1      1      1      1      1      1      1      1      1
4      1      1      1      1      1      1      1      1      1      1
5      1      1      1      1      1      1      1      1      1      1
6      1      1      1      1      1      1      1      1      1      1
7      1      1      1      1      1      1      1      1      1      1
8      1      1      1      1      1      1      1      1      1      1
9      1      1      1      1      1      1      1      1      1      1
10     1      1      1      1      1      1      1      1      1      1
11     1      1      1      1      1      1      1      1      1      1
12     1      1      1      1      1      1      1      1      1      1
13     1      1      1      1      1      1      1      1      1      1
14     1      1      1      1      1      1      1      1      1      1
15     1      1      1      1      1      1      1      1      1      1
;

param D2
:      1      2      3      4      5      6      7      8      :=
1      1      1      1      1      1      1      1      1
2      1      1      1      1      1      1      1      1
3      1      1      1      1      1      1      1      1
4      1      1      1      1      1      1      1      1
5      1      1      1      1      1      1      1      1
6      1      1      1      1      1      1      1      1
7      1      1      1      1      1      1      1      1
8      1      1      1      1      1      1      1      1
9      1      1      1      1      1      1      1      1
10     1      1      1      1      1      1      1      1
;
```

An hierarchical location-allocation model: Data

```
param D3
: 1 2 3 4 5 6 :=
1 1 1 1 1 1
2 1 1 1 1 1
3 1 1 1 1 1
4 1 1 1 1 1
5 1 1 1 1 1
6 1 1 1 1 1
7 1 1 1 1 1
8 1 1 1 1 1
;

# Travel cost/ patients h, from PHC i to IHC j
param TC1
: 1 2 3 4 5 6 7 8 9 10 :=
1 1 1 1 1 1 1 1 1 1
2 1 1 1 1 1 1 1 1 1
3 1 1 1 1 1 1 1 1 1
4 1 1 1 1 1 1 1 1 1
5 1 1 1 1 1 1 1 1 1
6 1 1 1 1 1 1 1 1 1
7 1 1 1 1 1 1 1 1 1
8 1 1 1 1 1 1 1 1 1
9 1 1 1 1 1 1 1 1 1
10 1 1 1 1 1 1 1 1 1
11 1 1 1 1 1 1 1 1 1
12 1 1 1 1 1 1 1 1 1
13 1 1 1 1 1 1 1 1 1
14 1 1 1 1 1 1 1 1 1
15 1 1 1 1 1 1 1 1 1
;

param TC2
: 1 2 3 4 5 6 7 8 :=
1 1 1 1 1 1 1 1
2 1 1 1 1 1 1 1
3 1 1 1 1 1 1 1
4 1 1 1 1 1 1 1
5 1 1 1 1 1 1 1
6 1 1 1 1 1 1 1
7 1 1 1 1 1 1 1
8 1 1 1 1 1 1 1
9 1 1 1 1 1 1 1
10 1 1 1 1 1 1 1
;
```

An hierarchical location-allocation model: Data

```
param TC3
:   1       2       3       4       5       6       :=
1   1       1       1       1       1       1
2   1       1       1       1       1       1
3   1       1       1       1       1       1
4   1       1       1       1       1       1
5   1       1       1       1       1       1
6   1       1       1       1       1       1
7   1       1       1       1       1       1
8   1       1       1       1       1       1
;

# Variable cost of IHC j / patient h
param VC1
:   1       2       3       4       5       6       7       8       9       10      :=
1   5       5       5       5       5       5       5       5       5       5
2   5       5       5       5       5       5       5       5       5       5
;

param VC2
:   1       2       3       4       5       6       7       8       :=
1   10      10      10      10      10      10      10      10
2   10      10      10      10      10      10      10      10
;

param VC3
:   1       2       3       4       5       6       :=
1   20      20      20      20      20      20
2   20      20      20      20      20      20
;

param FC1:=
1   1000
2   1000
3   1000
4   1000
5   1000
6   1000
7   100
8   100
9   100
10  100
;
```

An hierarchical location-allocation model: Data

```
param FC2:=  
1      200  
2      200  
3      200  
4      200  
5      200  
6      200  
7      200  
8      200  
;
```

```
param FC3:=  
1      300  
2      300  
3      300  
4      300  
5      300  
6      300  
;
```

```
# The population size of type p at point i  
param W  
:      1      2      :=  
1      100    200  
2      100    200  
3      100    200  
4      100    200  
5      100    200  
6      100    200  
7      100    200  
8      100    200  
9      100    200  
10     100    200  
11     100    200  
12     100    200  
13     100    200  
14     100    200  
15     100    200  
;
```

An hierarchical location-allocation model: Data

```
# Ministry of Health parameter for requirements PHC
param MS1:=
ME1      2e-2
EF1 2e-3
ACS 2e-3
DE1      2e-3
;

param MS2:=
ME2 1e-2
EF2 1e-2
FON 1e-2
OCD 1e-2
DE2 1e-2
;

param MS3:=
ME3 1e-2
EF3 2e-2
TE3 1e-2
FIS 2e-2
DE3      1e-2
;

param CNES1
:      1      2      3      4      5      6      7      8      9      10      :=
ME1    5      5      5      5      5      5      5      5      5      5
EF1    5      5      5      5      5      5      5      5      5      5
ACS    5      5      5      5      5      5      5      5      5      5
DE1    5      5      5      5      5      5      5      5      5      5
;

param CNES2
:      1      2      3      4      5      6      7      8      :=
ME2    3      3      3      3      3      3      3      3
EF2    3      3      3      3      3      3      3      3
FON    3      3      3      3      3      3      3      3
OCD    3      3      3      3      3      3      3      3
DE2    3      3      3      3      3      3      3      3
;
```

An hierarchical location-allocation model: Data

```
param CNES3
:      1      2      3      4      5      6      :=
ME3 2  2      2      2      2      2
EF3 2  2      2      2      2      2
TE3 2  2      2      2      2      2
FIS 2  2      2      2      2      2
DE3 2  2      2      2      2      2
;

# The capacity of a level-1 PCF in K.
param C1
:      1      2      3      4      5      6      7      8      9      10     :=
1      1000   1000   1000   1000   1000   1000   1000   500   500   500
2      1000   1000   1000   1000   1000   1000   1000   500   500   500
;

param C2:=
1      1000
2      100
3      100
4      200
5      200
6      200
7      200
8      200
;

param C3:=
1      300
2      300
3      300
4      300
5      300
6      300
;

# The number of a level-k to be established.
param U:=
1      4
2      3
3      2
;
```

An hierarchical location-allocation model: Data

```
# The proportion of patients in a L-1 to a L-2 param O1:=  
1      0.4  
2      0.4  
3      0.4  
4      0.4  
5      0.4  
6      0.4  
7      0.4  
8      0.4  
9      0.4  
10     0.4  
;  
  
param O2:=  
1      0.7  
2      0.7  
3      0.7  
4      0.7  
5      0.7  
6      0.7  
7      0.7  
8      0.7  
;  
  
end;
```

Outline of this lecture

Modelling exercise: Data

Modelling exercise: Report

Modelling exercise: Model

Modelling exercise: Code

An hierarchical location-allocation model: Report

```

=====
Health Care Plan
=====
Logist cost:      $7560.00
Fixed cost [E]:   $7700.00
Fixed cost [C]:   $1000.00
New team cost [C]: $500.00
Variable Cost:    $108900.00
=====
Total Cost:      $125660.00
=====
New Units: Qty  Max  Use (%)
=====
PHC      : 0    4    0.00%
SHC      : 2    3    66.67%
THC      : 2    2    100.00%
=====
Municipality:      Pop Flow
=====
[1      ]: 300  300
[2      ]: 300  300
[3      ]: 300  300
[4      ]: 300  300
[5      ]: 300  300
[6      ]: 300  300
[7      ]: 300  300
[8      ]: 300  300
[9      ]: 300  300
[10     ]: 300  300

```

```

[11     ]: 300  300
[12     ]: 300  300
[13     ]: 300  300
[14     ]: 300  300
[15     ]: 300  300
=====
Mun      > PHC : (flow)
=====
M[1     ] > : 300
          > L[1 ]: 300
M[2     ] > : 300
          > L[1 ]: 300
M[3     ] > : 300
          > L[2 ]: 300
M[4     ] > : 300
          > L[3 ]: 300
M[5     ] > : 300
          > L[3 ]: 300
M[6     ] > : 300
          > L[4 ]: 300
M[7     ] > : 300
          > L[5 ]: 300
M[8     ] > : 300
          > L[5 ]: 300
M[9     ] > : 300
          > L[1 ]: 300
M[10    ] > : 300
          > L[6 ]: 300
M[11    ] > : 300
          > L[5 ]: 300

```

```

M[12    ] > : 300
          > L[6 ]: 300
M[13    ] > : 300
          > L[6 ]: 300
M[14    ] > : 300
          > L[6 ]: 300
M[15    ] > : 300
          > L[1 ]: 300
=====
PHC      > SHC : (flow)
=====
L[1     ] > : 480
          > L[3 ]: 100
          > L[4 ]: 180
          > L[8 ]: 200
L[2     ] > : 120
          > L[5 ]: 120
L[3     ] > : 240
          > L[1 ]: 240
L[4     ] > : 120
          > L[1 ]: 120
L[5     ] > : 360
          > L[1 ]: 360
L[6     ] > : 480
          > L[1 ]: 280
          > L[2 ]: 100
          > L[4 ]: 20
          > L[5 ]: 80

```

An hierarchical location-allocation model: Report

```
=====
SHC    > THC    : (flow)
=====
L[1 ] > : 700
      > L[3 ]: 110
      > L[4 ]: 290
      > L[5 ]: 300
L[2 ] > : 70
      > L[1 ]: 70
L[3 ] > : 70
      > L[2 ]: 70
L[4 ] > : 140
      > L[3 ]: 140
L[5 ] > : 140
      > L[1 ]: 90
      > L[3 ]: 50
L[8 ] > : 140
      > L[1 ]: 140
=====
```

```
=====
Health care team (Existing and New*)
=====
PHC-Team CNES    Flow Lack/Excess
=====
L[1 ]
  [ME1]: 5.00    24.00 19.00
  [EF1]: 5.00    2.40 -2.60
  [ACS]: 5.00    2.40 -2.60
  [DE1]: 5.00    2.40 -2.60
L[2 ]
  [ME1]: 5.00    6.00  1.00
  [EF1]: 5.00    0.60 -4.40
  [ACS]: 5.00    0.60 -4.40
  [DE1]: 5.00    0.60 -4.40
L[3 ]
  [ME1]: 5.00    12.00 7.00
  [EF1]: 5.00    1.20 -3.80
  [ACS]: 5.00    1.20 -3.80
  [DE1]: 5.00    1.20 -3.80
L[4 ]
  [ME1]: 5.00    6.00  1.00
  [EF1]: 5.00    0.60 -4.40
  [ACS]: 5.00    0.60 -4.40
  [DE1]: 5.00    0.60 -4.40
L[5 ]
  [ME1]: 5.00    18.00 13.00
  [EF1]: 5.00    1.80 -3.20
  [ACS]: 5.00    1.80 -3.20
```

```
[DE1]: 5.00    1.80 -3.20
L[6 ]
  [ME1]: 5.00    24.00 19.00
  [EF1]: 5.00    2.40 -2.60
  [ACS]: 5.00    2.40 -2.60
  [DE1]: 5.00    2.40 -2.60
=====
SHC-Team CNES    Flow Lack/Excess
=====
L[1 ]
  [ME2]: 3.00    10.00 7.00
  [EF2]: 3.00    10.00 7.00
  [FON]: 3.00    10.00 7.00
  [TOC]: 3.00    10.00 7.00
  [DE2]: 3.00    10.00 7.00
L[2 ]
  [ME2]: 3.00    1.00 -2.00
  [EF2]: 3.00    1.00 -2.00
  [FON]: 3.00    1.00 -2.00
  [TOC]: 3.00    1.00 -2.00
  [DE2]: 3.00    1.00 -2.00
L[3 ]
  [ME2]: 3.00    1.00 -2.00
  [EF2]: 3.00    1.00 -2.00
  [FON]: 3.00    1.00 -2.00
  [TOC]: 3.00    1.00 -2.00
  [DE2]: 3.00    1.00 -2.00
```

An hierarchical location-allocation model: Report

L[4]				L[2]				[2] [2]: 1000 300 30%			
[ME2]: 3.00 2.00 -1.00				[ME3]: 2.00 0.70 -1.30				[3] [1]: 1000 0 0%			
[EF2]: 3.00 2.00 -1.00				[EF3]: 2.00 1.40 -0.60				[3] [2]: 1000 600 60%			
[FON]: 3.00 2.00 -1.00				[TE3]: 2.00 0.70 -1.30				[4] [1]: 1000 0 0%			
[TOC]: 3.00 2.00 -1.00				[FIS]: 2.00 1.40 -0.60				[4] [2]: 1000 300 30%			
[DE2]: 3.00 2.00 -1.00				[DE3]: 2.00 0.70 -1.30				[5] [1]: 1000 0 0%			
L[5*]				L[3]				[5] [2]: 1000 900 90%			
[ME2]: 2.00 2.00				[ME3]: 2.00 3.00 1.00				[6] [1]: 1000 200 20%			
[EF2]: 2.00 2.00				[EF3]: 2.00 6.00 4.00				[6] [2]: 1000 1000 100%			
[FON]: 2.00 2.00				[TE3]: 2.00 3.00 1.00				=====			
[TOC]: 2.00 2.00				[FIS]: 2.00 6.00 4.00				SHC : Capty Met Use (%)			
[DE2]: 2.00 2.00				[DE3]: 2.00 3.00 1.00				=====			
L[8*]				L[4*]				[1] : 1000 1000 100%			
[ME2]: 2.00 2.00				[ME3]: 2.90 2.90				[2] : 100 100 100%			
[EF2]: 2.00 2.00				[EF3]: 5.80 5.80				[3] : 100 100 100%			
[FON]: 2.00 2.00				[TE3]: 2.90 2.90				[4] : 200 200 100%			
[TOC]: 2.00 2.00				[FIS]: 5.80 5.80				[5 *]: 200 200 100%			
[DE2]: 2.00 2.00				[DE3]: 2.90 2.90				[8 *]: 200 200 100%			
=====				L[5*]				=====			
THC-Team CNES Flow Lack/Excess				[ME3]: 3.00 3.00				THC : Capty Met Use (%)			
=====				[EF3]: 6.00 6.00				=====			
L[1]				[TE3]: 3.00 3.00				[1] : 300 300 100%			
[ME3]: 2.00 3.00 1.00				[FIS]: 6.00 6.00				[2] : 300 70 23%			
[EF3]: 2.00 6.00 4.00				[DE3]: 3.00 3.00				[3] : 300 300 100%			
[TE3]: 2.00 3.00 1.00				=====				[4 *]: 300 290 97%			
[FIS]: 2.00 6.00 4.00				PHC [p]: Capty Met Use (%)				[5 *]: 300 300 100%			
[DE3]: 2.00 3.00 1.00				=====				=====			
				[1] [1]: 1000 200 20%							
				[1] [2]: 1000 1000 100%							
				[2] [1]: 1000 0 0%							

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Modelling exercise: Code

An hierarchical location-allocation model: Formulation

I : The set of demand points.

K : Health care levels (PHC, SHC, THC)

P : Types of Patients (chronic or acute)

E_k : Health care team from level k

EL_k : Existing health care units on three levels

CL_k : Candidate health care locations on three levels

$L_k = EL_k \cup CL_k$: ALL health care locations on three levels

An hierarchical location-allocation model: Formulation

I : The set of demand points.

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P : Types of Patients (chronic or acute)

E_k : Health care team from level k

EL_k : Existing health care units on three levels

CL_k : Candidate health care locations on three levels

$L_k = EL_k \cup CL_k$: ALL health care locations on three levels

- ▶ CE_c^1 : Team cost $c \in E_1$ from PHC level (\$/month).
- ▶ CE_c^2 : Team cost $c \in E_2$ from SHC level (\$/month).
- ▶ CE_c^3 : Team cost $c \in E_3$ from THC level (\$/month).

An hierarchical location-allocation model: Formulation

I : The set of demand points.

K : Health care levels (PHC, SHC, THC)

P : Types of Patients (chronic or acute)

E_k : Health care team from level k

EL_k : Existing health care units on three levels

CL_k : Candidate health care locations on three levels

$L_k = EL_k \cup CL_k$: ALL health care locations on three levels

- ▶ CE_c^1 : Team cost $c \in E_1$ from PHC level (\$/month).
- ▶ CE_c^2 : Team cost $c \in E_2$ from SHC level (\$/month).
- ▶ CE_c^3 : Team cost $c \in E_3$ from THC level (\$/month).
- ▶ D_{ij}^1 : The travel time between patient origin $i \in I$ and PHC $j \in L_1$ (min).
- ▶ $D_{jj'}^2$: The travel time between PHC location $j \in L_1$ and SHC $j' \in L_2$ (min).
- ▶ $D_{j'j*}^3$: The travel time between SHC location $j' \in L_2$ and THC $j* \in L_3$ (min).

An hierarchical location-allocation model: Formulation

- ▶ TC_{ij}^1 : The travel cost between patient origin $i \in I$ and PHC $j \in L_1$ (\$/min).
- ▶ $TC_{jj'}^2$: The travel cost between PHC location $j \in L_1$ and SHC $j' \in L_2$ (\$/min).
- ▶ $TC_{j'j*}^3$: The travel cost between SHC location $j' \in L_2$ and THC $j* \in L_3$ (\$/min).

An hierarchical location-allocation model: Formulation

- ▶ TC_{ij}^1 : The travel cost between patient origin $i \in I$ and PHC $j \in L_1$ (\$/min).
- ▶ $TC_{jj'}^2$: The travel cost between PHC location $j \in L_1$ and SHC $j' \in L_2$ (\$/min).
- ▶ $TC_{j'j*}^3$: The travel cost between SHC location $j' \in L_2$ and THC $j* \in L_3$ (\$/min).
- ▶ FC_j^1 : Fixed cost per period for operating the PHC unit at $j \in L_1$ (\$/month).
- ▶ FC_j^2 : Fixed cost per period for operating the SHC unit at $j' \in L_2$ (\$/month).
- ▶ FC_j^3 : Fixed cost per period for operating the THC unit at $j* \in L_3$ (\$/month).

An hierarchical location-allocation model: Formulation

- ▶ TC_{ij}^1 : The travel cost between patient origin $i \in I$ and PHC $j \in L_1$ (\$/min).
- ▶ $TC_{jj'}^2$: The travel cost between PHC location $j \in L_1$ and SHC $j' \in L_2$ (\$/min).
- ▶ $TC_{j'j*}^3$: The travel cost between SHC location $j' \in L_2$ and THC $j* \in L_3$ (\$/min).
- ▶ FC_j^1 : Fixed cost per period for operating the PHC unit at $j \in L_1$ (\$/month).
- ▶ FC_j^2 : Fixed cost per period for operating the SHC unit at $j' \in L_2$ (\$/month).
- ▶ FC_j^3 : Fixed cost per period for operating the THC unit at $j* \in L_3$ (\$/month).
- ▶ W_{ip} : The population size of type $p \in P$ at demand point $i \in I$ (pop).

An hierarchical location-allocation model: Formulation

- ▶ TC_{ij}^1 : The travel cost between patient origin $i \in I$ and PHC $j \in L_1$ (\$/min).
- ▶ $TC_{jj'}^2$: The travel cost between PHC location $j \in L_1$ and SHC $j' \in L_2$ (\$/min).
- ▶ $TC_{j'j*}^3$: The travel cost between SHC location $j' \in L_2$ and THC $j* \in L_3$ (\$/min).
- ▶ FC_j^1 : Fixed cost per period for operating the PHC unit at $j \in L_1$ (\$/month).
- ▶ FC_j^2 : Fixed cost per period for operating the SHC unit at $j' \in L_2$ (\$/month).
- ▶ FC_j^3 : Fixed cost per period for operating the THC unit at $j* \in L_3$ (\$/month).
- ▶ W_{ip} : The population size of type $p \in P$ at demand point $i \in I$ (pop).
- ▶ MS_e^1 : Ministry of Health: Requirements for PHC staff $e \in E_1$ (prof/pop).
- ▶ MS_e^2 : Ministry of Health: Requirements for SHC staff $e \in E_2$ (prof/pop).
- ▶ MS_e^3 : Ministry of Health: Requirements for THC staff $e \in E_3$ (prof/pop).

An hierarchical location-allocation model: Formulation

- ▶ $CNES_{ej}^1$: PHC current staff $e \in E_1$ at PHC unit $j \in EL_1$ (prof).
- ▶ $CNES_{ej'}^2$: SHC current staff $e \in E_2$ at SHC unit $j' \in EL_2$ (prof).
- ▶ $CNES_{ej*}^3$: THC current staff $e \in E_3$ at THC unit $j* \in EL_3$ (prof).

An hierarchical location-allocation model: Formulation

- ▶ $CNES_{ej}^1$: PHC current staff $e \in E_1$ at PHC unit $j \in EL_1$ (prof).
- ▶ $CNES_{ej'}^2$: SHC current staff $e \in E_2$ at SHC unit $j' \in EL_2$ (prof).
- ▶ $CNES_{ej*}^3$: THC current staff $e \in E_3$ at THC unit $j* \in EL_3$ (prof).
- ▶ C_{pj}^1 : The capacity for population $p \in P$ at PHC unit $j \in L_1$ (pop).
- ▶ $C_{j'}^2$: The capacity at SHC unit $j' \in L_2$ (pop).
- ▶ C_{j*}^3 : The capacity at THC unit $j* \in L_3$ (pop).

An hierarchical location-allocation model: Formulation

- ▶ $CNES_{ej}^1$: PHC current staff $e \in E_1$ at PHC unit $j \in EL_1$ (prof).
- ▶ $CNES_{ej'}^2$: SHC current staff $e \in E_2$ at SHC unit $j' \in EL_2$ (prof).
- ▶ $CNES_{ej*}^3$: THC current staff $e \in E_3$ at THC unit $j* \in EL_3$ (prof).
- ▶ C_{pj}^1 : The capacity for population $p \in P$ at PHC unit $j \in L_1$ (pop).
- ▶ $C_{j'}^2$: The capacity at SHC unit $j' \in L_2$ (pop).
- ▶ C_{j*}^3 : The capacity at THC unit $j* \in L_3$ (pop).
- ▶ U_k : The number of additional units to be created at level $k \in K$ (unit).

An hierarchical location-allocation model: Formulation

- ▶ $CNES_{ej}^1$: PHC current staff $e \in E_1$ at PHC unit $j \in EL_1$ (prof).
- ▶ $CNES_{ej'}^2$: SHC current staff $e \in E_2$ at SHC unit $j' \in EL_2$ (prof).
- ▶ $CNES_{ej*}^3$: THC current staff $e \in E_3$ at THC unit $j* \in EL_3$ (prof).
- ▶ C_{pj}^1 : The capacity for population $p \in P$ at PHC unit $j \in L_1$ (pop).
- ▶ $C_{j'}^2$: The capacity at SHC unit $j' \in L_2$ (pop).
- ▶ C_{j*}^3 : The capacity at THC unit $j* \in L_3$ (pop).
- ▶ U_k : The number of additional units to be created at level $k \in K$ (unit).
- ▶ O_j^1 : The proportion of patients referred from $j \in L_1$ to $j' \in L_2$ (%).
- ▶ $O_{j'}^2$: The proportion of patients referred from $j' \in L_2$ to $j* \in L_3$ (%).

An hierarchical location-allocation model: Formulation

- ▶ $y_{ij} \in \{0, 1\}$: If population $i \in I$ is allocated to PHC unit at $j \in L_1$ (1) or not (0)
- ▶ $y_j^1 \in \{0, 1\}$: If the PHC unit located at $j \in L_1$ is used (1) or not (0)
- ▶ $y_{j'}^2 \in \{0, 1\}$: If the SHC unit located at $j' \in L_2$ is used (1) or not (0)
- ▶ $y_{j*}^3 \in \{0, 1\}$: If the THC unit located at $j* \in L_3$ is used (1) or not (0)
- ▶ u_{pij}^1 : Patients $p \in P$ flow between origin $i \in I$ and PHC unit at $j \in L_1$ (pop).
- ▶ $u_{jj'}^2$: The flow between the PHC unit at $j \in L_1$ and SHC unit at $j' \in L_2$ (pop).
- ▶ $u_{j'j*}^3$: The flow between the SHC unit at $j' \in L_2$ and THC unit at $j* \in L_3$ (pop).
- ▶ l_{ej}^1 : Lack (or excess) of staff of type $e \in E_1$ at location $j \in L_1$ (prof)
- ▶ $l_{ej'}^2$: Lack (or excess) of staff of type $e \in E_2$ at location $j' \in L_2$ (prof)
- ▶ l_{ej*}^3 : Lack (or excess) of staff of type $e \in E_3$ at location $j* \in L_3$ (prof)

An hierarchical location-allocation model: Formulation

$$\min. \sum_p^P \sum_i^I \sum_j^{L_1} D_{ij}^1 TC_{ij}^1 u_{pij}^1 + \sum_j^{L_1} \sum_{j'}^{L_2} D_{jj'}^2 TC_{jj'}^2 u_{jj'}^2 + \sum_{j'}^{L_2} \sum_{j*}^{L_3} D_{j'j*}^3 TC_{j'j*}^3 u_{j'j*}^3 +$$

An hierarchical location-allocation model: Formulation

$$\begin{aligned} \min. \quad & \sum_p^P \sum_i^I \sum_j^{L_1} D_{ij}^1 TC_{ij}^1 u_{pij}^1 + \sum_j^{L_1} \sum_{j'}^{L_2} D_{jj'}^2 TC_{jj'}^2 u_{jj'}^2 + \sum_{j'}^{L_2} \sum_{j*}^{L_3} D_{j'j*}^3 TC_{j'j*}^3 u_{j'j*}^3 + \\ & \sum_j^{EL_1} FC_j^1 y_j^1 + \sum_{j'}^{EL_2} FC_{j'}^2 y_{j'}^2 + \sum_{j*}^{EL_3} FC_{j*}^3 y_{j*}^3 + \\ & \sum_j^{CL_1} FC_j^1 y_j^1 + \sum_{j'}^{CL_2} FC_{j'}^2 y_{j'}^2 + \sum_{j*}^{CL_3} FC_{j*}^3 y_{j*}^3 + \end{aligned}$$

An hierarchical location-allocation model: Formulation

$$\min. \sum_p^P \sum_i^I \sum_j^{L_1} D_{ij}^1 TC_{ij}^1 u_{pij}^1 + \sum_j^{L_1} \sum_{j'}^{L_2} D_{jj'}^2 TC_{jj'}^2 u_{jj'}^2 + \sum_{j'}^{L_2} \sum_{j*}^{L_3} D_{j'j*}^3 TC_{j'j*}^3 u_{j'j*}^3 +$$

$$\sum_j^{EL_1} FC_j^1 y_j^1 + \sum_{j'}^{EL_2} FC_{j'}^2 y_{j'}^2 + \sum_{j*}^{EL_3} FC_{j*}^3 y_{j*}^3 +$$

$$\sum_j^{CL_1} FC_j^1 y_j^1 + \sum_{j'}^{CL_2} FC_{j'}^2 y_{j'}^2 + \sum_{j*}^{CL_3} FC_{j*}^3 y_{j*}^3 +$$

$$\sum_j^{CL_1} \sum_c^{E_1} CE_c^1 y_j^1 + \sum_{j'}^{CL_2} \sum_c^{E_2} CE_c^2 y_{j'}^2 + \sum_{j*}^{CL_3} \sum_c^{E_3} CE_c^3 y_{j*}^3 +$$

An hierarchical location-allocation model: Formulation

$$\begin{aligned}
 \min. \quad & \sum_p^P \sum_i^I \sum_j^{L_1} D_{ij}^1 TC_{ij}^1 u_{pij}^1 + \sum_j^{L_1} \sum_{j'}^{L_2} D_{jj'}^2 TC_{jj'}^2 u_{jj'}^2 + \sum_{j'}^{L_2} \sum_{j*}^{L_3} D_{j'j*}^3 TC_{j'j*}^3 u_{j'j*}^3 + \\
 & \sum_j^{EL_1} FC_j^1 y_j^1 + \sum_{j'}^{EL_2} FC_{j'}^2 y_{j'}^2 + \sum_{j*}^{EL_3} FC_{j*}^3 y_{j*}^3 + \\
 & \sum_j^{CL_1} FC_j^1 y_j^1 + \sum_{j'}^{CL_2} FC_{j'}^2 y_{j'}^2 + \sum_{j*}^{CL_3} FC_{j*}^3 y_{j*}^3 + \\
 & \sum_j^{CL_1} \sum_c^{E_1} CE_c^1 y_j^1 + \sum_{j'}^{CL_2} \sum_c^{E_2} CE_c^2 y_{j'}^2 + \sum_{j*}^{CL_3} \sum_c^{E_3} CE_c^3 y_{j*}^3 + \\
 & \sum_p^P \sum_i^I \sum_j^{L_1} VC_{ij}^1 u_{pij}^1 + \sum_j^{L_1} \sum_{j'}^{L_2} VC_{jj'}^2 u_{jj'}^2 + \sum_{j'}^{L_2} \sum_{j*}^{L_3} VC_{j'j*}^3 u_{j'j*}^3
 \end{aligned} \tag{1}$$

An hierarchical location-allocation model: Formulation

- ▶ We fix the location variables of **existing** all health care units

An hierarchical location-allocation model: Formulation

- We fix the location variables of **existing** all health care units

$$\text{s.t.: } y_j^1 = 1 \quad \forall j \in EL_1 \quad (2)$$

$$y_{j'}^2 = 1 \quad \forall j' \in EL_2 \quad (3)$$

$$y_{j*}^3 = 1 \quad \forall j* \in EL_3 \quad (4)$$

An hierarchical location-allocation model: Formulation

- ▶ We fix the location variables of **existing** all health care units

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- ▶ Universal coverage: The entire population at each demand point i must be assigned to (an existing or candidate) **closest** PHC location. (Unit: pop = pop)

An hierarchical location-allocation model: Formulation

- We fix the location variables of **existing** all health care units

$$\text{s.t.: } y_j^1 = 1 \quad \forall j \in EL_1 \quad (2)$$

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$$y_{j*}^3 = 1 \quad \forall j* \in EL_3 \quad (4)$$

- Universal coverage: The entire population at each demand point i must be assigned to (an existing or candidate) **closest** PHC location. (Unit: pop = pop)

$$\sum_p^P W_{ip} y_{ij} = \sum_p^P u_{pij}^1 \quad \forall i \in I, j \in L_1 \quad (5)$$

$$\sum_j^{L_1} y_{ij} = 1 \quad \forall i \in I \quad (6)$$

$$\sum_{k|D_{ik}^1 > D_{ij}^1}^{L_1} y_{ik} + y_j^1 \leq 1 \quad \forall i \in I, j \in L_1 \quad (7)$$

An hierarchical location-allocation model: Formulation

- ▶ Then, population assigned to PHC can be referred to SHC and THC. ($pop = pop$)

An hierarchical location-allocation model: Formulation

- Then, population assigned to PHC can be referred to SHC and THC. (pop = pop)

$$\sum_{j'}^{L_2} u_{jj'}^2 = O_j^1 \sum_p^P \sum_i^I u_{pij}^1 \quad \forall j \in L_1 \quad (8)$$

$$\sum_{j^*}^{L_3} u_{j'j^*}^3 = O_{j'}^2 \sum_j^{L_1} u_{jj'}^2 \quad \forall j' \in L_2 \quad (9)$$

An hierarchical location-allocation model: Formulation

- ▶ Then, population assigned to PHC can be referred to SHC and THC. (pop = pop)

$$\sum_{j'}^{L_2} u_{jj'}^2 = O_j^1 \sum_p^P \sum_i^I u_{pij}^1 \quad \forall j \in L_1 \quad (8)$$

$$\sum_{j^*}^{L_3} u_{j'j^*}^3 = O_{j'}^2 \sum_j^{L_1} u_{jj'}^2 \quad \forall j' \in L_2 \quad (9)$$

- ▶ Optimize existing location health care team (Unit: pop*(prof/pop) - prof = prof)

An hierarchical location-allocation model: Formulation

- Then, population assigned to PHC can be **referred** to SHC and THC. (pop = pop)

$$\sum_{j'}^{L_2} u_{jj'}^2 = O_j^1 \sum_p^P \sum_i^I u_{pij}^1 \quad \forall j \in L_1 \quad (8)$$

$$\sum_{j^*}^{L_3} u_{j'j^*}^3 = O_{j'}^2 \sum_j^{L_1} u_{jj'}^2 \quad \forall j' \in L_2 \quad (9)$$

- Optimize **existing location** health care team (Unit: pop*(prof/pop) - prof = prof)

$$\sum_p^P \sum_i^I u_{pij}^1 MS_e^1 - l_{ej}^1 = CNES_{ej}^1 \quad \forall j \in EL_1, e \in E_1 \quad (10)$$

$$\sum_j^{EL_1} u_{jj'}^2 MS_e^2 - l_{ej'}^2 = CNES_{ej'}^2 \quad \forall j' \in EL_2, e \in E_2 \quad (11)$$

$$\sum_{j'}^{EL_2} u_{j'j^*}^3 MS_e^3 - l_{ej^*}^3 = CNES_{ej^*}^3 \quad \forall j^* \in EL_3, e \in E_3 \quad (12)$$

An hierarchical location-allocation model: Formulation

- ▶ Optimize **candidate location** future health care team (Unit: prof = prof)

An hierarchical location-allocation model: Formulation

- Optimize **candidate location** future health care team (Unit: prof = prof)

$$\sum_p^P \sum_i^I u_{pij}^1 MS_e^1 = l_{ej}^1 \quad \forall j \in CL_1, e \in E_1 \quad (13)$$

$$\sum_j^{L_1} u_{jj'}^2 MS_e^2 = l_{ej'}^2 \quad \forall j' \in CL_2, e \in E_2 \quad (14)$$

$$\sum_{j'}^{L_2} u_{j'j*}^3 MS_e^3 = l_{ej*}^3 \quad \forall j* \in CL_3, e \in E_3 \quad (15)$$

An hierarchical location-allocation model: Formulation

- ▶ Existing health care units have limited capacity (Unit: pop = pop)

An hierarchical location-allocation model: Formulation

- Existing health care units have limited capacity (Unit: pop = pop)

$$\sum_i^I u_{pij}^1 \leq C_{pj}^1 \quad \forall j \in EL_1, p \in P \quad (16)$$

$$\sum_j^{L_1} u_{jj'}^2 \leq C_{j'}^2 \quad \forall j' \in EL_2 \quad (17)$$

$$\sum_{j'}^{L_2} u_{j'j*}^3 \leq C_{j*}^3 \quad \forall j* \in EL_3 \quad (18)$$

An hierarchical location-allocation model: Formulation

- ▶ Creation of **new** *capacitated* health care units? (Unit: pop = pop)

An hierarchical location-allocation model: Formulation

- Creation of **new** *capacitated* health care units? (Unit: pop = pop)

$$\sum_i^I u_{pij}^1 \leq C_{pj}^1 y_j^1 \quad \forall j \in L_1, p \in P \quad (19)$$

$$\sum_j^{L_1} u_{jj'}^2 \leq C_{j'}^2 y_{j'}^2 \quad \forall j' \in L_2 \quad (20)$$

$$\sum_{j'}^{L_2} u_{j'j*}^3 \leq C_{j*}^3 y_{j*}^3 \quad \forall j* \in L_3 \quad (21)$$

An hierarchical location-allocation model: Formulation

- ▶ Creation of new health care units is limited by a budget (units = units)

An hierarchical location-allocation model: Formulation

- ▶ Creation of new health care units is limited by a budget (units = units)

$$\sum_j^{L_1} y_j^1 \leq U_1 \quad (22)$$

$$\sum_{j'}^{L_2} y_{j'}^2 \leq U_2 \quad (23)$$

$$\sum_{j^*}^{L_3} y_{j^*}^3 \leq U_3 \quad (24)$$

An hierarchical location-allocation model: Formulation

► Domain of the model variables

$$y_{ij} \in \{0, 1\} \quad \forall i \in I, j \in L_1 \quad (25)$$

$$y_j^1 \in \{0, 1\} \quad \forall j \in L_1 \quad (26)$$

$$y_{j'}^2 \in \{0, 1\} \quad \forall j' \in L_2 \quad (27)$$

$$y_{j*}^3 \in \{0, 1\} \quad \forall j* \in L_3 \quad (28)$$

$$u_{pij}^1 \geq 0 \quad \forall p \in P, i \in I, j \in L_1 \quad (29)$$

$$u_{jj'}^2 \geq 0 \quad \forall j \in L_1, j' \in L_2 \quad (30)$$

$$u_{j'j*}^3 \geq 0 \quad \forall j' \in L_2, j* \in L_3 \quad (31)$$

Outline of this lecture

Modelling exercise: Data

Modelling exercise: Report

Modelling exercise: Model

Modelling exercise: Code

An hierarchical location-allocation model: Code

```
1 #####
2 # Operations Research for Health Care
3 # Lesson 10/15
4 # Author: João Flávio de Freitas Almeida <joao.flavio@dep.ufmg.br>
5 # LIPOLIV: Laboratório de Estudos em Planejamento de Operações Integradas
6 # Departamento de Engenharia de Produção
7 # Universidade Federal de Minas Gerais - Escola de Engenharia
8 #####
9 # Health Care Facility Location Problem: Considering fixed facilities,
10 # choose intermediate facilities according a criteria that improve service
11 # quality. Consider health care teams.
12 #####
13 # gplsol -m hc.mod -d hc.dat --cuts
14
15 set I; # The set of demand points.
16
17 set K; # Health care levels (PHC, SHC, THC)
18
19 set P; # Types of Patients (chronic or acute)
20
21 set E[K]; # Health care team from level k
22
23 set EL[K]; # EXISTING health care units on three levels
24
25 set CL[K]; # CANDIDATE health care LOCATIONS on three levels
26
27 set L[k in K] := EL[k] union CL[k];
28
29 #####
30 param C1[c1 in E[1]]; # Team cost K1 ($/month)
31 param C2[c2 in E[2]]; # Team cost K2 ($/month)
32 param C3[c3 in E[3]]; # Team cost K3 ($/month)
33
34 param D1[I,1]; # The travel time between i and level-1 PCF. (min)
35 param D2[L[1],L[2]]; # The travel time between candidate L1 and L2. (min)
36 param D3[L[2],L[3]]; # The travel time between candidate L2 and L3. (min)
37
38 param TC1[I,1]; # Travel cost/pat p, from demand point i to L1 ($/min)
39 param TC2[L[1],L[2]]; # Travel cost/pat p, from L1 to L2 ($/min)
40 param TC3[L[2],L[3]]; # Travel cost/pat p, from L2 to L3 ($/min)
41
42 param VC1[P,1]; # Variable cost of PHC j / pop h ($/pop)
43 param VC2[P,2]; # Variable cost of SHC j / pop h ($/pop)
44 param VC3[P,3]; # Variable cost of THC j / pop h ($/pop)
45
46 param FC1[L[1]]; # Fixed cost per period for operating PHC j ($/month)
47 param FC2[L[2]]; # Fixed cost per period for operating SHC j ($/month)
48 param FC3[L[3]]; # Fixed cost per period for operating THC j ($/month)
49
50 #####
51
52 param W[I,P]; # The population size of type p at demand point i (pop)
53
54 param MS1[E[1]]; # Ministry of Health parameter for requirements PHC (prof/pop)
55 param MS2[E[2]]; # Ministry of Health parameter for requirements SHC (prof/pop)
56 param MS3[E[3]]; # Ministry of Health parameter for requirements THC (prof/pop)
57
58 param CNES1[E[1],L[1]]; # Health professional teams PHC at location L1 (prof)
59 param CNES2[E[2],L[2]]; # Health professional teams PHC at location L2 (prof)
60 param CNES3[E[3],L[3]]; # Health professional teams PHC at location L3 (prof)
61
62 # Service operating capacity at IHC j
63 param C1[P,L[1]]; # The capacity of a level-1 PCF in K. (pop)
64 param C2[L[2]]; # The capacity of a level-2 PCF in J. (pop)
65 param C3[L[3]]; # The capacity of a level-3 PCF in J. (pop)
66
67 param U[K]; # The number of UNITS level-k to be established. (unit)
68
69 param O1[L[1]]; # The proportion of patients in a L-1 to a L-2 PCF. (%)
70 param O2[L[2]]; # The proportion of patients in a L-1 to a L-2 PCF. (%)
71
72 #####
73 var y1[i in I, j1 in L[1]], >=0, binary; # 1, if Pop is ASSIGNED to L-1 PCF (1) or not (0)
74 var y2[j2 in L[2]], >=0, binary; # 1, if a L-1 PCF is used (1) at loc. k. or not (0)
75 var y3[j3 in L[3]], >=0, binary; # 1, if a L-2 PCF is used (1) at loc. k. or not (0)
76
77 var u1[P, i in I, j1 in L[1]], >=0; # The flow p between demand point i and L1 (pop)
78 var u2[j1 in L[1], j2 in L[2]], >=0; # The flow between L1 and L2 (pop)
79 var u3[j2 in L[2], j3 in L[3]], >=0; # The flow between L2 and L3 (pop)
80
81 var l1[E[1],L[1]]; # Lack (or excess) of professional e on location L1 (prof)
82 var l2[E[2],L[2]]; # Lack (or excess) of professional e on location L2 (prof)
83 var l3[E[3],L[3]]; # Lack (or excess) of professional e on location L3 (prof)
84
85 #####
86
87
88
89
90
91
92
93
94
95
96
97
```

An hierarchical location-allocation model: Code

```
hmod
97
98 #####
99 # Minimizes social and business costs:
100 # the total demand-weighted travel distance (or time) +
101 # the total fixed costs of operating the facilities
102 # (related to health care teams for each location) +
103 # the total variable costs / patient of operating the facilities.
104
105 minimize Total_costs:
106     # Patient transportation cost
107     sum(p in P, i in I, j1 in L[1]) D[i, j1] * TC[i, j1] * u1[p, i, j1]
108     + sum(j1 in L[1], j2 in L[2]) D[j1, j2] * TC[j1, j2] * u2[j1, j2]
109     + sum(j2 in L[2], j3 in L[3]) D[j2, j3] * TC[j2, j3] * u3[j2, j3]
110     # Cost of existing unit (including staff)
111     + sum(j1 in EL[1]) FC[i, j1] * y1[j1]
112     + sum(j2 in EL[2]) FC[j2] * y2[j2]
113     + sum(j3 in EL[3]) FC[j3] * y3[j3]
114     # New unit cost
115     + sum(j1 in CL[1]) FC[i, j1] * y1[j1]
116     + sum(j2 in CL[2]) FC[j2] * y2[j2]
117     + sum(j3 in CL[3]) FC[j3] * y3[j3]
118     # Cost of new staff
119     + sum(j1 in CL[1], c1 in E[1]) CE[c1] * y1[j1]
120     + sum(j2 in CL[2], c2 in E[2]) CE[c2] * y2[j2]
121     + sum(j3 in CL[3], c3 in E[3]) CE[c3] * y3[j3]
122     # Variable cost per patient
123     + sum(p in P, i in I, j1 in L[1]) D[i, j1] * VC[i, p, j1] * u1[p, i, j1]
124     + sum(p in P, j1 in L[1], j2 in L[2]) D[j1, j2] * VC[p, j2] * u2[j1, j2]
125     + sum(p in P, j2 in L[2], j3 in L[3]) D[j2, j3] * VC[p, j3] * u3[j2, j3];
126
127 # Fix variables of EXISTING location
128 s.t. F1[j1 in EL[1]]: y1[j1] = 1;
129 s.t. F2[j2 in EL[2]]: y2[j2] = 1;
130 s.t. F3[j3 in EL[3]]: y3[j3] = 1;
131
132 # Entire population at each demand point i must be assigned
133 # to (existing and candidate) location L1
134 # (pop) = (pop)
135 s.t. R0[i in I, j1 in L[1]]: sum(p in P) u1[p, i, j1] = sum(p in P) u1[p, i, j1];
136 s.t. R0a[i in I]: sum(j1 in L[1]) y1[j1] = 1;
137
138 # Flow balance from PHC > SHC > THC
139 # (pop) = (pop)
140 s.t. R1[j1 in L[1]]: sum(j2 in L[2]) u2[j1, j2] = 01[j1] * sum(p in P, i in I) u1[p, i, j1];
141 s.t. R2[j2 in L[2]]: sum(j3 in L[3]) u3[j2, j3] = 02[j2] * sum(j1 in L[1]) u2[j1, j2];
142
143 # Team of existing
144 # (pop) * (prof/pop) - (prof) = (prof)
145 # (prof) = (prof)
146 s.t. R3a[j1 in EL[1], e1 in E[1]]: sum(p in P, i in I) u1[p, i, j1] * MS1[e1] - 11[e1, j1] = CNE51[e1, j1];
147 s.t. R4a[j2 in EL[2], e2 in E[2]]: sum(j1 in L[1]) u2[j1, j2] * MS2[e2] - 12[e2, j2] = CNE52[e2, j2];
148 s.t. R5a[j3 in EL[3], e3 in E[3]]: sum(j2 in L[2]) u3[j2, j3] * MS3[e3] - 13[e3, j3] = CNE53[e3, j3];
149
150 # New team
151 # (prof) = (prof)
152 s.t. R3c[j1 in CL[1], e1 in E[1]]: sum(p in P, i in I) u1[p, i, j1] * MS1[e1] - 11[e1, j1] = CNE51[e1, j1];
153 s.t. R4c[j2 in CL[2], e2 in E[2]]: sum(j1 in L[1]) u2[j1, j2] * MS2[e2] - 12[e2, j2] = CNE52[e2, j2];
154 s.t. R5c[j3 in CL[3], e3 in E[3]]: sum(j2 in L[2]) u3[j2, j3] * MS3[e3] - 13[e3, j3] = CNE53[e3, j3];
155
156 # Capacity of existing (patients)
157 # (pop) = (pop)
158 s.t. R6a[j1 in EL[1], p in P]: sum(i in I) u1[p, i, j1] <= C1[p, j1];
159 s.t. R7a[j2 in EL[2]]: sum(j1 in L[1]) u2[j1, j2] <= C2[j2];
160 s.t. R8a[j3 in EL[3]]: sum(j2 in L[2]) u3[j2, j3] <= C3[j3];
161
162 # Activation of new units (?)
163 # (pop) = (pop)
164 s.t. R6c[j1 in CL[1], p in P]: sum(i in I) u1[p, i, j1] <= C1[p, j1] * y1[j1];
165 s.t. R7c[j2 in CL[2]]: sum(j1 in L[1]) u2[j1, j2] <= C2[j2] * y2[j2];
166 s.t. R8c[j3 in CL[3]]: sum(j2 in L[2]) u3[j2, j3] <= C3[j3] * y3[j3];
167
168 # The number of alev-k to be established.
169 # (units) = (units)
170 s.t. R9c: sum(j1 in CL[1]) y1[j1] <= U[1];
171 s.t. R10c: sum(j2 in CL[2]) y2[j2] <= U[2];
172 s.t. R11c: sum(j3 in CL[3]) y3[j3] <= U[3];
173
174
175 solve;
```

An hierarchical location-allocation model: Code

[illegible]

An hierarchical location-allocation model: Code

```
# hcmold
269 }
270 }
271 printf: "-----\n";
272 printf: "SHC-Team CNES\\tFlow\\tLack\\tExcess\\n";
273 printf: "-----\n";
274 for{j2 in EL[2]: sum{j1 in L[1]}u2[j1,j2] > 0}{
275   printf"L[X-2d]\\n", j2;
276   for{e2 in E[2]}{
277     printf" [X-s]: %2f\\t%2f\\t%2f\\n", e2, CNES2[e2,j2],
278     sum{j1 in L[1]}u2[j1,j2]*MS2[e2],
279     l2[e2,j2];
280   }
281 }
282 for{j2 in CL[2]: sum{j1 in L[1]}u2[j1,j2] > 0}{
283   printf"L[X-d*]\\n", j2;
284   for{e2 in E[2]: sum{j1 in L[1]}u2[j1,j2]>0}{
285     printf" [X-s]: \\t%2f\\t%2f\\n", e2,
286     sum{j1 in L[1]}u2[j1,j2]*MS2[e2],
287     l2[e2,j2];
288   }
289 }
290 printf: "-----\n";
291 printf: "THC-Team CNES\\tFlow\\tLack\\tExcess\\n";
292 printf: "-----\n";
293 for{j3 in EL[3]: sum{j2 in L[2]}u3[j2,j3] > 0}{
294   printf"L[X-2d]\\n", j3;
295   for{e3 in E[3]}{
296     printf" [X-s]: %2f\\t%2f\\t%2f\\n", e3, CNES3[e3,j3],
297     sum{j2 in L[2]}u3[j2,j3]*MS3[e3],
298     l3[e3,j3];
299   }
300 }
301 for{j3 in CL[3]: sum{j2 in L[2]}u3[j2,j3] > 0}{
302   printf"L[X-d*]\\n", j3;
303   for{e3 in E[3]: sum{j2 in L[2]}u3[j2,j3]>0}{
304     printf" [X-s]: \\t%2f\\t%2f\\n", e3,
305     sum{j2 in L[2]}u3[j2,j3]*MS3[e3],
306     l3[e3,j3];
307   }
308 }
309 }
310 printf: "-----\n";
311 printf: "PHC [p]:\\tCapt\\tMet\\tUse(XX)\\n";
312 printf: "-----\n";
313 # Existing location
314 printf{j1 in EL[1], p in P}:
315 "[X-3d][Xd]:\\tXd\\tXd\\tXd\\n", j1, p,
316 C1[p,j1],
317 sum{i in I}u1[p,i,j1],
318 ((sum{i in I}u1[p,i,j1])/(C1[p,j1]))*100;
319 # Candidate location
320 printf{j1 in CL[1], p in P: sum{i in I}u1[p,i,j1]>0}:
321 "[X-2d*][Xd]:\\tXd\\tXd\\tXd\\n", j1, p,
322 C1[p,j1],
323 sum{i in I}u1[p,i,j1],
324 ((sum{i in I}u1[p,i,j1])/(C1[p,j1]))*100;
325 }
326 printf: "-----\n";
327 printf: "SHC :\\tCapt\\tMet\\tUse(XX)\\n";
328 printf: "-----\n";
329 # Existing location
330 printf{j2 in EL[2]: "[X-6d]:\\tXd\\tXd\\tXd\\n", j2,
331 C2[j2],
332 sum{j1 in L[1]}u2[j1,j2],
333 ((sum{j1 in L[1]}u2[j1,j2])/(C2[j2]))*100;
334 # Candidate location
335 printf{j2 in CL[2]: sum{j1 in L[1]}u2[j1,j2]>0}:
336 "[X-5d*]:\\tXd\\tXd\\tXd\\n", j2,
337 C2[j2],
338 sum{j1 in L[1]}u2[j1,j2],
339 ((sum{j1 in L[1]}u2[j1,j2])/(C2[j2]))*100;
340 }
341 printf: "-----\n";
342 printf: "THC :\\tCapt\\tMet\\tUse(XX)\\n";
343 printf: "-----\n";
344 # Existing location
345 printf{j3 in EL[3]: "[X-6d]:\\tXd\\tXd\\tXd\\n", j3,
346 C3[j3],
347 sum{j2 in L[2]}u3[j2,j3],
348 ((sum{j2 in L[2]}u3[j2,j3])/(C3[j3]))*100;
349 # Candidate location
350 printf{j3 in CL[3]: sum{j2 in L[2]}u3[j2,j3]>0}:
351 "[X-5d*]:\\tXd\\tXd\\tXd\\n", j3,
352 C3[j3],
353 sum{j2 in L[2]}u3[j2,j3],
354 ((sum{j2 in L[2]}u3[j2,j3])/(C3[j3]))*100;
355 printf: "-----\n\\n";
356 end;
357
```